

Thermodynamic State Vector Interface & Nonequilibrium Exergy Accounting in Planetary Biogeochemical Monads

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Sprint 018

Repository: <https://github.com/pascalranoroarijaona/WebOfLife>

Abstract

As planetary-scale simulators advance to model complex ecophysiological metabolism and coupled biogeochemical cycles (Carbon, Nitrogen, Phosphorus, and Water), rigorous tracking of conserved quantities and irreversible dissipation channels is paramount. Sprint 018 establishes the **Thermodynamic State Vector Interface** within the *Web of Life* framework (`src/thermodynamics/types.ts` and `src/thermodynamics/thermodynamic_monad_process.ts`). This architecture enforces strict mathematical contracts for internal entropy generation ($\dot{S}_{\text{gen}} \geq 0$), exergy destruction rate via the Gouy-Stodola theorem ($\dot{I} = T_0 \dot{S}_{\text{gen}}$), and boundary flux arrays. By casting biological monads as open thermodynamic control volumes, the system guarantees adherence to the First and Second Laws of Thermodynamics across all simulation epochs.

1 Introduction & Systems Ecology Motivation

Complex ecological systems operate far from thermodynamic equilibrium, maintaining internal organization and macroscopic homeostasis by dissipating high-grade solar exergy and exporting entropy to space. Traditional ecological models frequently track mass and energy balance independently of thermodynamic driving potentials.

To bridge systems ecology and nonequilibrium thermodynamics, Sprint 018 formalizes state vector interfaces that track intensive/extensive thermodynamic properties and boundary fluxes for elemental stocks $\mathbf{M} = [M_C, M_N, M_P, M_{H_2O}]^T$.

2 Mathematical Framework

2.1 First Law: Mass-Energy Conservation

For any open thermodynamic control volume V bounded by surface ∂V , the internal energy E evolves according to boundary heat transfers, useful work, and mass-enthalpy transport:

$$\frac{dE}{dt} = \sum_i \dot{Q}_i - \dot{W} + \sum_j \dot{m}_j \left(h_j + \frac{v_j^2}{2} + gz_j \right) \quad (1)$$

2.2 Second Law: Nonequilibrium Entropy Production

The temporal evolution of system entropy S accounts for boundary entropy exchanges and internal irreversibilities:

$$\frac{dS}{dt} = \sum_i \frac{\dot{Q}_i}{T_i} + \sum_j \dot{m}_j s_j + \dot{S}_{\text{gen}} \quad (2)$$

where the Second Law requires non-negative internal entropy generation:

$$\dot{S}_{\text{gen}} \geq 0 \quad \forall t \quad (3)$$

2.3 Gouy-Stodola Theorem & Exergy Destruction

Exergy destruction ($\dot{I}$) is quantified relative to the ambient dead-state temperature T_0 :

$$\dot{I} = T_0 \dot{S}_{\text{gen}} \geq 0 \quad (4)$$

3 Implementation Architecture

The core TypeScript interfaces implemented in `src/thermodynamics/types.ts` enforce invariant checking and immutable state transitions across simulation steps. The monad process verifies mass conservation within tolerance $\epsilon = 10^{-12}$ and asserts non-negative entropy generation prior to committing state updates.

4 Conclusion & Future Directions

Sprint 018 successfully establishes type-safe thermodynamic accounting within the *Web of Life* engine. Future sprints will integrate multi-pod exergy coupling networks and spatial transport equations to model planetary biosphere resilience under anthropogenic forcing.

Official Repository: <https://github.com/pascalranoroarijaona/WebOfLife>